E_1-chiral quantum groups and the E_N ladder

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1 The Yangian as E_1-chiral quantum group

The Yangian $Y_\hbar(\mathfrak{sl}_2)$ is presented by generators $t_{ij}^{(r)}$, $1 \le i,j \le 2$, $r \ge 1$, assembled into the generating series $T(u) = \mathbf{1} + \sum_{r\ge1} T^{(r)} u^{-r}$ with $T^{(r)} = (t_{ij}^{(r)})$, subject to the RTT relation

$$R(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R(u-v), \qquad R(z) = 1 - \frac{\hbar\,P}{z}, \tag{1}$$

where $P \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)$ is the permutation.

Multiplication. The RTT relation (1) defines a $\mathbb{Z}$-filtered associative algebra. It is *not* commutative: $[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)})$. Multiplication is E_1 and no more.

Coproduct. The spectral coproduct

$$\Delta_z \colon Y_\hbar(\mathfrak{sl}_2) \to Y_\hbar(\mathfrak{sl}_2) \,\widehat{\otimes}\, Y_\hbar(\mathfrak{sl}_2), \qquad \Delta_z(T(u)) = T(u) \otimes T(u-z), \tag{2}$$

is coassociative: $(\mathrm{id}\widehat{\otimes}\,\Delta_w) \circ \Delta_z = (\Delta_z \widehat{\otimes}\,\mathrm{id}) \circ \Delta_{z+w}$. At $z = 0$ it reduces to the Drinfeld coproduct $\Delta_0(T(u)) = T(u) \otimes T(u)$.

R-matrix and quasi-triangularity. The Yang R-matrix $R(z) = 1 - \hbar P/z$ satisfies the Yang–Baxter equation

$$R_{12}(z_1 - z_2)\,R_{13}(z_1 - z_3)\,R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)\,R_{13}(z_1 - z_3)\,R_{12}(z_1 - z_2). \tag{3}$$

Quasi-triangularity: $R(z)\,\Delta_z(a) = \Delta_z^{\mathrm{op}}(a)\,R(z)$ for all $a \in Y_\hbar(\mathfrak{sl}_2)$.

Braided module category. For $Y_\hbar(\mathfrak{sl}_2)$-modules M, N, the spectral coproduct (2) defines a fusion product $M \otimes_z N$ by pulling back the $(Y_\hbar \widehat{\otimes} Y_\hbar)$-action. The braiding

$$\beta_{M,N}(z) \colon M \otimes_z N \xrightarrow{R(z)\cdot\sigma} N \otimes_z M$$

satisfies the braid relation by (3).

What makes $Y_\hbar(\mathfrak{sl}_2)$ genuinely E_1. The RTT relation (1) is *not* derived from a local OPE on a curve. There is no vertex algebra V for which $Y_\hbar(\mathfrak{sl}_2)$ arises as a chiral envelope. The ordering of tensor factors is independent input. In the hierarchy

$$E_1\text{-chiral} \supsetneq E_\infty\text{-chiral (vertex algebras)},$$

the Yangian sits in $E_1 \setminus E_\infty$: genuinely nonlocal.

This example earns the following definition.

2 E_1-chiral quantum groups

Definition 2.1. *An E_1-**chiral quantum group** on a smooth curve X is a tuple $(\mathcal{A},\, \mu,\, \Delta,\, R,\, S)$ where:*

(i) $\mathcal{A}$ is an E_1-chiral algebra on X: an E_1-algebra object in factorisation algebras on X;

(ii) μ is the E_1-chiral multiplication (the A_∞-structure $\{m_k\}_{k\geq 2}$ encoding the ordered OPE);

(iii) $\Delta\colon \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ is a chiral coproduct (coassociative coalgebra structure on $\mathcal{A}$ itself, in the chiral tensor category);

(iv) $R(z) \in \mathcal{A}^{\widehat{\otimes} 2}$ is a spectral R-matrix satisfying the Yang–Baxter equation (3)*;*

(v) $R(z)\,\Delta(a) = \Delta^{\mathrm{op}}(a)\,R(z)$ for all $a \in \mathcal{A}$ (quasi-triangularity);

(vi) $S\colon \mathcal{A} \to \mathcal{A}$ is an antipode satisfying $\mu \circ (\mathrm{id} \otimes S) \circ \Delta = \eta \circ \varepsilon$ and its mirror.

The coproduct Δ is structure on $\mathcal{A}$. It is not the deconcatenation coproduct on the bar complex $B(\mathcal{A})$: deconcatenation is a structural feature of any bar construction and carries no information about $\mathcal{A}$ beyond what the product already encodes.

Theorem 2.2 (Categorical level shift)**.** *Let $(\mathcal{A}, \mu, \Delta, R, S)$ be an E_1-chiral quantum group. Then:*

(a) *$\mathcal{A}$ is an E_1-algebra.*

(b) $\mathrm{Mod}_{\mathcal{A}}$ *is an E_2-algebra object in* Cat*: it is a braided monoidal category with braiding $\beta_{M,N}(z) = R(z) \cdot \sigma$.*

(c) *The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{RHom}_{\mathcal{A}\text{-}\mathcal{A}}(\mathcal{A}, \mathcal{A})$ carries an E_2-chiral structure.*

E_2 lives on $\mathrm{Mod}_{\mathcal{A}}$ and on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, never on $\mathcal{A}$.

Proof. (a) is the definition.

(b) The coproduct Δ defines the monoidal product $M \otimes_\Delta N$ on $\mathrm{Mod}_{\mathcal{A}}$ by restriction of scalars along Δ. Coassociativity of Δ gives associativity of $\otimes_\Delta$. The R-matrix $R(z)$ and quasi-triangularity give the braiding natural transformation $\beta_{M,N}\colon M \otimes N \to N \otimes M$ satisfying the hexagon axioms (which follow from the YBE (3)). The antipode S gives duals.

(c) The Deligne conjecture (Kontsevich, Tamarkin): the Hochschild cochain complex of an E_1-algebra carries an E_2-algebra structure. The chiral variant: for an E_1-chiral algebra, the chiral Hochschild cochains $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ carry E_2-chiral structure, computing the chiral derived centre. □

3 The bar complex: what it sees and what it does not

Let $\mathcal{A}$ be an E_1-chiral algebra on a smooth curve X.

Proposition 3.1 (Bar complex as E_1 coalgebra [1, Theorem A])**.** *The bar construction*

$$B(\mathcal{A}) = \bigoplus_{n\geq 0} (s\mathcal{A})^{\otimes n}, \qquad d_B = \sum_{k\geq 2}\sum_i m_k^{(i)}, \qquad \Delta(a_1|\cdots|a_n) = \sum_{i=0}^{n} (a_1|\cdots|a_i)\otimes(a_{i+1}|\cdots|a_n),$$

is an E_1 dg coassociative coalgebra on $\mathrm{Ran}(X)$*. The differential d_B is a coderivation of Δ. The cobar construction Ω provides a left adjoint: $\Omega B(\mathcal{A}) \simeq \mathcal{A}$ for augmented $\mathcal{A}$.*

The bar complex computes $B(\mathcal{A}) \simeq k \otimes^{\mathbf{L}}_{\mathcal{A}} k$, the factorisation homology of the interval $[0,1]$ with augmentation boundary conditions. It is not factorisation homology of the line: $\int_{\mathbb{R}} \mathcal{A} \simeq \mathcal{A}$.

Proposition 3.2 (Classical r-matrix from collision residues)**.** *Let $\mathcal{A} = V_k(\mathfrak{g})$ be the Kac–Moody vertex algebra at level k. The genus-0, degree-2 collision residue of the bar differential gives*

$$r(z) = \frac{k\,\Omega}{z}, \tag{4}$$

where $\Omega \in \mathfrak{g}\otimes\mathfrak{g}$ is the Casimir. This $r(z)$ satisfies the classical Yang–Baxter equation.

Proof. The m_2 component of d_B encodes the singular OPE $J^a(z)J^b(w) \sim k\,\kappa^{ab}/(z-w) + f^{ab}_c J^c(w)/(z-w)$. The $d\log$-extraction procedure replaces the OPE pole $1/(z-w)$ by the logarithmic form $d\log(z-w)$. The collision residue at $z=w$ in degree 2 extracts the coefficient $k\,\Omega = k\sum_a J^a \otimes J_a$, divided by the spectral parameter $z = z_1 - z_2$. The CYBE follows from the Jacobi identity of $\mathfrak{g}$ (equivalently, from the codimension-2 stratum of $\mathrm{FM}_3(\mathbb{C})$, which imposes the three-term relation). □

Proposition 3.3 (Curvature)**.** *At genus $g \geq 1$, the bar complex has*

$$d_B^2 = \kappa_{\mathrm{ch}} \cdot \omega_g, \tag{5}$$

where κ_{ch} is the shadow obstruction class and ω_g is the Arakelov $(1,1)$-form on the genus-g surface, normalised so that $\int_{\Sigma_g}\omega_g = 1$.

The bar complex sees the *product* (fusion, encoded in d_B) and the *classical r-matrix* (from collision residues).

It does *not* see:

(i) the chiral coproduct $\Delta\colon \mathcal{A} \to \mathcal{A}\otimes\mathcal{A}$ (the coproduct encodes splitting, not fusion; it is independent algebraic structure);

(ii) the full quantum R-matrix $R(z)$ (only its classical limit $r(z)$ appears in the shadow tower);

(iii) quasi-triangularity or the antipode.

The shadow obstruction tower $\{F_g\}_{g\geq 1}$, extracted from (5), detects collision data. The coproduct encodes the opposite operation and is invisible to the tower.

Sources of the chiral coproduct: Gaiotto–Rapčák–Zhou [4] construct coproducts on deformed double current algebras of $\mathfrak{gl}_K$ from M2–M5 fusions; Jindal–Kaubrys–Latyntsev [5] construct vertex coproducts on CoHAs of quivers with potential, recovering Drinfeld's deformed coproduct for ADE quivers.

4 The E_N ladder

For an E_1-chiral algebra $\mathcal{A}$ on X, the following structures are obtained by ascending the E_N hierarchy. Each step adds one E_1 factor via Dunn additivity.

E_1: the boundary algebra

$\mathcal{A}$ is E_1-chiral: an E_1-algebra in factorisation algebras on X. The bar complex $B(\mathcal{A})$ is the E_1 dg coassociative coalgebra of Proposition 3.1.

Status: proved (Proposition 3.1).

$E_1 \to E_2$: Hochschild and Deligne

Theorem 4.1 (E_2 on the derived centre)**.** *The chiral derived centre*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$$

carries an E_2-chiral algebra structure. The E_2 decomposition $E_2 \simeq E_1^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}$ (Dunn additivity) gives:

(a) E_1^{hol}*: the chiral algebra structure of $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ on X;*

(b) E_1^{top}*: the A_∞-algebra structure in the transverse direction.*

The pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the Swiss-cheese datum: the bulk E_2 acts on the boundary E_1.

Status: proved. The chiral Deligne conjecture follows from Kontsevich formality and homotopy transfer to the chiral setting. Dunn additivity applies to $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, not to $\mathcal{A}$.

$E_2 \to E_3$-chiral: the holomorphic-topological bulk

An E_3-chiral structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is the factorisation algebra of a 3d holomorphic-topological theory on $X \times \mathbb{R}$:

$$E_3 \simeq E_2^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}.$$

This requires a 3d HT theory whose boundary condition is $\mathcal{A}$.

Theorem 4.2 (E_3-chiral for Kac–Moody [7])**.** *For $\mathcal{A} = V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, holomorphic Chern–Simons theory on $X \times \mathbb{R}$ provides the 3d HT bulk. The factorisation algebra of observables on $X \times \mathbb{R}$ is an E_3-chiral algebra, and its restriction to the boundary $X \times \{0\}$ recovers $V_k(\mathfrak{g})$.*

Status: proved for $\mathcal{A} = V_k(\mathfrak{g})$ by Costello–Li and Costello–Francis–Gwilliam. For a general vertex algebra $\mathcal{A}$, the existence of a 3d HT theory with boundary $\mathcal{A}$ is open.

E_3-chiral $\to$ E_3-topological: topologisation

Theorem 4.3 (Topologisation for Kac–Moody)**.** *Let $\mathcal{A} = V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$. Write $T(z) = \{Q, G(z)\}$ where $T(z)$ is the Sugawara conformal vector, Q is the BRST operator, and $G(z)$ is the antighost. Then:*

(a) *Translations along X are Q-exact: the complex structure of X is irrelevant in Q-cohomology.*

(b) *The E_3-chiral structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ (Theorem 4.2) descends to an E_3-topological structure on $H^\bullet(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), Q)$.*

(c) *The resulting E_3-topological theory is Chern–Simons at level $k + h^\vee$.*

Status: proved [6, 7]. The E_3-topological factorisation homology trace reproduces the Reshetikhin–Turaev link invariant.

Conjecture 4.4 (Topologisation for general chiral algebras)**.** *Let $\mathcal{A}$ be an E_1-chiral algebra carrying a conformal vector $T(z)$ at non-critical level, and suppose $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ admits an E_3-chiral structure from a 3d holomorphic-topological bulk. Then $T(z) = \{Q, G(z)\}$ implies that the E_3-chiral structure descends to E_3-topological on $H^\bullet(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), Q)$.*

The conjecture presupposes the existence of the E_3-chiral structure, which is itself open for general $\mathcal{A}$ (see the discussion after Theorem 4.2).

At the *critical level* $k = -h^\vee$, the Sugawara construction degenerates. No conformal vector exists. The E_3-chiral structure persists but topologisation fails: the theory is stuck at $\mathrm{SC}^{\mathrm{ch,top}}$, the Swiss-cheese structure with two distinct colours (holomorphic and topological).

Remark 4.5 (The generic case is $\mathrm{SC}^{\mathrm{ch,top}}$, not E_3). E_3-topological requires a conformal vector. Most chiral algebras (Heisenberg, lattice VOAs, critical-level Kac–Moody, free fields without Sugawara) lack one or have it only at special levels. The Swiss-cheese structure $\mathrm{SC}^{\mathrm{ch,top}}$, in which the two colours remain distinct, is the generic structure on $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$.

5 The missing operad and open problems

The $E_2 \to E_3$ step for general $\mathcal{A}$ (not necessarily Kac–Moody) requires constructing an A_∞-algebra in E_1-chiral algebras. The operad governing such objects has not been constructed.

Conjecture 5.1 (The chiral-A_∞ modular operad)**.** *There exists a modular operad $\mathcal{O}^{\text{ch-}A_\infty}$ whose algebras consist of:*

(a) *an E_1-chiral algebra structure on X (governed by $\mathrm{SC}^{\mathrm{ch,top}}$ on the within-surface strata);*

(b) *an A_∞-algebra structure in a transverse direction $\mathbb{R}$;*

(c) *compatibility: the modular operad encodes how transverse A_∞-operations interact with the E_1 ordering on X;*

(d) *at genus $g \geq 1$: curvature $\kappa_{\mathrm{ch}} \cdot \omega_g$ as the obstruction to extending the genus-0 structure.*

The genus-0 part of $\mathcal{O}^{\text{ch-}A_\infty}$ recovers $\mathrm{SC}^{\mathrm{ch,top}}$.

Conjecture 5.2 (Curved Dunn additivity)**.** *For a curved E_2-algebra with curvature $d^2 = \kappa\cdot\omega$, there exists a decomposition as a curved E_1-algebra in curved E_1-algebras, with $\kappa \cdot \omega$ as the obstruction to the flat (uncurved) Dunn decomposition $E_2 \simeq E_1 \otimes E_1$.*

Question 5.3. *Does the Gaiotto–Rapčák–Zhou coproduct [4] (type A, from M2–M5 fusions) or the Jindal–Kaubrys–Latyntsev vertex coproduct [5] (ADE quivers, from CoHA) lift to a chiral coproduct on $V_k(\mathfrak{g})$ for arbitrary simple $\mathfrak{g}$?*

Question 5.4. *Construct an E_1-chiral quantum group $\mathcal{A}$ that is not E_∞: an E_1-chiral algebra with chiral coproduct Δ, R-matrix $R(z)$, and antipode S making $\mathrm{Mod}_{\mathcal{A}}$ braided monoidal, where the E_1 ordering is not derived from any vertex algebra OPE. The Yangian $Y_\hbar(\mathfrak{sl}_2)$ of §1 is the candidate; the missing ingredient is a rigorous construction of the chiral algebra structure on a curve X (as opposed to the formal-variable presentation).*

Question 5.5. *The E_2 braiding from the Deligne conjecture (Theorem 4.1) is categorical, with no spectral parameter. The R-matrix $R(z)$ of a chiral quantum group carries a spectral parameter z from the holomorphic direction. Exhibit the natural isomorphism between the categorical braiding on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ and the R-matrix braiding $R(z)$ on $\mathrm{Mod}_{\mathcal{A}}$, in the $\mathcal{D}$-module enriched setting where position-dependence arises from the $\mathcal{D}$-module structure.*

Question 5.6. *The boundary-linear bulk reconstruction $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \simeq C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is proved. Extend this to the full global triangle: boundary $\to$ bulk $\to$ boundary. This is open beyond the Koszul locus.*

6 The equivalence triangle for $\mathcal{W}_{1+\infty}$

The preceding sections establish the E_1-chiral quantum group definition (Definition 2.1) and note that the coproduct Δ is not visible in the shadow tower. For the Yangian $Y_\hbar(\mathfrak{sl}_2)$, all three data (multiplication, coproduct, R-matrix) are supplied by the RTT formalism. The question is whether the same package can be assembled for class M algebras, where the bar-cobar adjunction does not produce a quasi-isomorphism at chains.

The test case is $\mathcal{W}_{1+\infty}$ at the self-dual level $c = 1$, where the Maulik–Okounkov (MO) R-matrix $R^{\mathrm{MO}}(u)$ is proved [1, 3]. Strategy 3: use R^{MO} as input and reconstruct the remaining data (A_∞ operations and coproduct) via the *equivalence triangle*

$$\begin{tikzcd} & A_\infty\text{-structure } \{m_k\} \arrow[dr, "\beta"] & \\ R\text{-matrix } R(z) \arrow[ur, "\alpha"] \arrow[rr, "\gamma^{-1}=\beta\circ\alpha"'] & & \text{coproduct } \Delta \end{tikzcd} \tag{6}$$

whose three functors are:

(a) α: *R-matrix* $\to A_\infty$. Restrict the spectral R-matrix data from $\mathrm{FM}^{\mathrm{ord}}_n(\mathbb{C})$ to the Stasheff associahedra $K_n \hookrightarrow \mathrm{FM}^{\mathrm{ord}}_n(\mathbb{C})$. The boundary strata of K_n encode the A_∞ operations m_k: K_n is the Stasheff compactification of the moduli of n points on $\mathbb{R}$ (a codimension-0 submanifold of $\mathrm{FM}^{\mathrm{ord}}_n(\mathbb{C})$), and the boundary stratification $\partial K_n = \bigcup_{p+q=n+1} K_p \times K_q$ recovers the A_∞ associativity relations $\sum m_p \circ m_q = 0$.

(b) β: $A_\infty \to$ *coproduct.* Given the A_∞ operations $\{m_k\}$, form the ordered bar coalgebra $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with bar differential $d_B = \sum m_k^{(i)}$. The cobar construction $\Omega(B^{\mathrm{ord}}(\mathcal{A}))$ produces an A_∞ algebra; dualizing the deconcatenation coproduct on B^{ord} through Koszul duality gives the twisted coproduct $\Delta_z\colon \mathcal{A}^!_{\mathrm{line}} \to \mathcal{A}^!_{\mathrm{line}} \widehat{\otimes} \mathcal{A}^!_{\mathrm{line}}$ on the open-colour Koszul dual.

(c) $\gamma^{-1} = \beta \circ \alpha$: *R-matrix* $\to$ *coproduct* (composite). The MO R-matrix $R^{\mathrm{MO}}(u)$ produces the coproduct directly, bypassing the intermediate A_∞ structure.

Step α: extracting A_∞ from the MO R-matrix

The structure function of $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}, \qquad h_1+h_2+h_3 = 0. \tag{7}$$

The MO R-matrix on the charge-n Fock space $\mathcal{F}_n$ has diagonal eigenvalues $R_{\lambda\lambda}(u) = \prod_{s\in\lambda} g(u+c(s))$ where $c(s) = h_1 i + h_2 j$ is the content of box $s = (i,j)$ in partition λ [3].

The functor α restricts the $\mathrm{FM}_n(\mathbb{C})$ configuration-space data to associahedra:

Construction 6.1 (α: R-matrix to A_∞)**.** *At arity n, the A_∞ operation m_n^{ch} is extracted from the R-matrix as*

$$m_n^{\mathrm{ch}}(a_1,\dots,a_n) \;=\; \int_{K_n} \omega_n^R(z_1,\dots,z_n;a_1,\dots,a_n), \tag{8}$$

where ω_n^R is the propagator form on K_n twisted by the R-matrix:

$$\omega_n^R \;=\; \bigwedge_{i<j} d\log(z_i - z_j) \;\cdot\; R_{i,i+1}(z_i - z_{i+1}) \;\cdot\; a_1(z_1)\otimes\cdots\otimes a_n(z_n).$$

For $n=2$: K_2 is a point, and m_2 is the binary product $m_2(a,b) = \mathrm{Res}_{z=0}\, a(z)\, b(0)\, d\log(z)$, the standard OPE extraction.

For $n=3$: $K_3 \simeq [0,1]$ is an interval. The integral over K_3 extracts the first higher product m_3^{ch} from the triple collision with the R-matrix twist:

$$m_3^{\mathrm{ch}}(a_1,a_2,a_3) \;=\; \int_0^1 \Big[\mathrm{Res}_{z_{12}=0}\; R_{12}(z_{12})\, a_1(z_1)\, a_2(z_2)\Big] \;\cdot\; a_3(z_3)\; d\log(z_{23})\, dt, \tag{9}$$

where the integration variable t parametrises K_3 and z_{12}, z_{23} are the successive differences. For class L (Kac–Moody, $R(z) = 1+\hbar\, P/z$), the integral evaluates to zero: $m_3 = 0$, consistent with Koszulness.

For $\mathcal{W}_{1+\infty}$ (class M): $m_3^{\mathrm{ch}} \neq 0$. The structure function (7) contributes terms at every order in $1/z$, producing a nonvanishing K_3 integral. In the spin-2 channel (Virasoro at $c=1$ inside $\mathcal{W}_{1+\infty}$), the m_3 operation encodes the first higher correction to the Virasoro OPE, consistent with $d_{\mathrm{gen}}(\mathrm{Vir}) = 3$ (generating depth 3: m_3 generates all higher m_k by recursive composition).

Step β: cobar dualization to the coproduct

Construction 6.2 (β: A_∞ to coproduct)**.** *Given the A_∞ operations $\{m_k^{\mathrm{ch}}\}$ from Construction 6.1, form the ordered bar complex*

$$B^{\mathrm{ord}}(\mathcal{W}_{1+\infty}) \;=\; T^c(s^{-1}\overline{\mathcal{W}_{1+\infty}})$$

with bar differential $d_B = \sum_{k\geq 2} m_k^{(i)}$ and deconcatenation coproduct. The open-colour Koszul dual is

$$\mathcal{A}^!_{\mathrm{line}} \;=\; H^*\big(B^{\mathrm{ord}}(\mathcal{W}_{1+\infty})\big)^\vee,$$

which is $Y(\widehat{\mathfrak{gl}}_1)$ (the affine Yangian of $\mathfrak{gl}_1$). The twisted coproduct on $\mathcal{A}^!_{\mathrm{line}}$ is obtained by dualizing the deconcatenation:

$$\Delta_z(f)(\bar a_1 \otimes \cdots \otimes \bar a_n) \;=\; \sum_{p=0}^{n} f(\bar a_1,\ldots,\bar a_p)\;\otimes_z\; f(\bar a_{p+1},\ldots,\bar a_n), \tag{10}$$

where $\otimes_z$ denotes the completed tensor product with spectral parameter z tracking the position of the cut. The twisting by z arises from the holomorphic coordinate: cutting the ordered sequence at position p separates the configuration into two clusters whose relative position is z.

The composite γ^{-1}: direct R-matrix to coproduct

Proposition 6.3 (Cohomological coproduct from MO R-matrix)**.** *At the cohomological level, the composite $\gamma^{-1} = \beta\circ\alpha$ applied to the MO R-matrix $R^{\mathrm{MO}}(u)$ of (7) produces a well-defined coproduct*

$$\Delta_z^{\mathrm{MO}}\colon Y(\widehat{\mathfrak{gl}}_1) \;\longrightarrow\; Y(\widehat{\mathfrak{gl}}_1)\,\widehat{\otimes}\,Y(\widehat{\mathfrak{gl}}_1) \tag{11}$$

satisfying:

(i) Coassociativity at cohomology: $(\Delta_z\widehat{\otimes}\,\mathrm{id})\circ\Delta_{z+w} = (\mathrm{id}\widehat{\otimes}\,\Delta_w)\circ\Delta_z$.

(ii) Quasi-triangularity at cohomology: $R^{\mathrm{MO}}(z)\,\Delta_z(a) = \Delta_z^{\mathrm{op}}(a)\,R^{\mathrm{MO}}(z)$ *for all* $a\in Y(\widehat{\mathfrak{gl}}_1)$.

(iii) Low-spin explicit form: *on the Cartan generators ψ_k (spin 1):*

$$\Delta_z(\psi(u)) \;=\; \psi(u)\otimes\psi(u-z),$$

where $\psi(u) = 1+\sum_{k\geq 0}\psi_k\,u^{-k-1}$; on the positive root generators e_k (spin 2):

$$\Delta_z(e(u)) \;=\; e(u)\otimes 1 \;+\; \psi(u)\otimes e(u-z);$$

and on the negative root generators f_k:

$$\Delta_z(f(u)) \;=\; 1\otimes f(u) \;+\; f(u-z)\otimes\psi(u).$$

These match the Schiffmann–Vasserot spectral coproduct at $z=0$ and the Drinfeld new coproduct Δ_{new} at leading spectral order.

Proof sketch. The well-definedness at cohomology follows from the general principle: the bar-cobar adjunction $\Omega \circ B \simeq \mathrm{id}$ holds at the level of the *coderived category* for all shadow classes, including class M [1, thm:bv-bar-coderived]. The obstruction to a chain-level quasi-isomorphism (which exists for classes G, L, C but fails for class M) does not affect the cohomological coproduct, because:

(a) The MO R-matrix is a cohomological object: it acts on $K_T(\mathrm{Hilb}^n) = H^*(B^{\mathrm{ord}})^\vee$, which is the cohomology of the bar complex, not the chain-level complex itself.

(b) The functor α (restriction to associahedra) commutes with passage to cohomology: the $K_n \hookrightarrow \mathrm{FM}_n^{\mathrm{ord}}(\mathbb{C})$ embedding respects the stratification that defines the bar differential.

(c) The functor β (Koszul dualization of deconcatenation) on cohomology is the classical Koszul duality, which is well-defined for any algebra (the coderived correction is invisible at cohomology).

Coassociativity (i) follows from the tautological coassociativity of deconcatenation on T^c, which survives dualization. Quasi-triangularity (ii) follows from the YBE for R^{MO} (proved in [3]).

The explicit formulas (iii) are computed as follows. The Cartan generator $\psi(u)$ is the diagonal part of the R-matrix action on the charge-1 Fock space. On $\mathcal{F}_1 = \bigoplus_\lambda \mathbb{C} \cdot |\lambda\rangle$ (partitions of charge 1, i.e. a single box), the R-matrix eigenvalue is $g(u) = (u - h_1)(u - h_2)(u - h_3)/((u + h_1)(u + h_2)(u + h_3))$. The Gauss decomposition $R = F^{-1} \cdot D \cdot E$ (lower triangular, diagonal, upper triangular) extracts the Drinfeld-type generators. The coproduct is read off from the matrix coproduct $\Delta(T(u)) = T(u) \dot{\otimes} T(u-z)$ applied to the Gauss factors. This is the Schiffmann–Vasserot coproduct for $Y(\widehat{\mathfrak{gl}}_1)$, obtained here from the MO R-matrix via γ^{-1}. □

Remark 6.4 (The class M obstruction). For class L algebras (Kac–Moody), γ^{-1} produces a coproduct at chains: the bar-cobar adjunction is a quasi-isomorphism ($\Omega(B(A)) \xrightarrow{\sim} A$), so all three vertices of the triangle (6) are equivalent at the chain level.

For class M algebras ($\mathcal{W}_{1+\infty}$, Virasoro), the chain-level bar-cobar map is *not* a quasi-isomorphism: the cobar $\Omega(B(A))$ is a "thickening" of A carrying additional acyclic data. The triangle (6) holds at:

(a) *Cohomology:* the coproduct (11) is well-defined on $H^*(B^{\mathrm{ord}})^\vee = Y(\widehat{\mathfrak{gl}}_1)$.

(b) *Coderived category:* the bar-cobar adjunction $\Omega \circ B \simeq \mathrm{id}$ holds in the coderived sense (see Vol I, thm:bv-bar-coderived), producing a coproduct on the coderived model of $\mathcal{W}_{1+\infty}$.

(c) *Chains (open):* whether γ^{-1} lifts to a chain-level chiral coproduct on the A_∞ model of $\mathcal{W}_{1+\infty}$ is a question about the formality of the A_∞ structure extracted by α. For class M, the A_∞ operations m_k ($k \geq 3$) are all nonzero, and the chain-level coproduct would need to be compatible with the full A_∞ tower. This is the content of Conjecture 6.5 below.

Conjecture 6.5 (Chain-level coproduct for class M)**.** *For $\mathcal{W}_{1+\infty}$ at $c = 1$, the cohomological coproduct Δ_z^{MO} of Proposition 6.3 lifts to a chain-level A_∞-coproduct $\Delta_z^{A_\infty} \colon \mathcal{W}_{1+\infty}^{A_\infty} \to$*

$\mathcal{W}_{1+\infty}^{A_\infty} \widehat{\otimes} \mathcal{W}_{1+\infty}^{A_\infty}$ *on the* A_∞ *model of* $\mathcal{W}_{1+\infty}$ *obtained from Construction 6.1, making* $(\mathcal{W}_{1+\infty}^{A_\infty}, \Delta_z^{A_\infty}, R^{\mathrm{MO}}(z))$ *an* A_∞*-bialgebra in the sense of Markl–Shnider.*

Evidence: the Jindal–Kaubrys–Latyntsev vertex coproduct on the CoHA of $\mathbb{C}^3$ *[5] provides a chain-level construction for the positive half* $Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathrm{CoHA}(\mathbb{C}^3)$*; the question is whether this extends through the Drinfeld double to the full affine Yangian.*

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